



JANUARY 1

Healthcare Landscape Analysis in Kenya

Correlating Hospital Availability, Disease Burden and Mortality Trends in Kenya (2010–2015)



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Business Problem

Problem Statement



Despite efforts to improve healthcare access, disparities in morbidity rates and hospital availability persist across counties.

This project explores how limited infrastructure may contribute to higher disease burden, aiming to identify high-risk counties for targeted health investment.



Questions we seek to answer

How does hospital availability at the county level correlate with disease incidence, and can national mortality data offer context for health challenges despite the lack of direct county-level links?



Data Overview



Data Details

Data Source : Kenya National Bureau of Statistics (KNBS)

Methods of data cleaning & transformation:

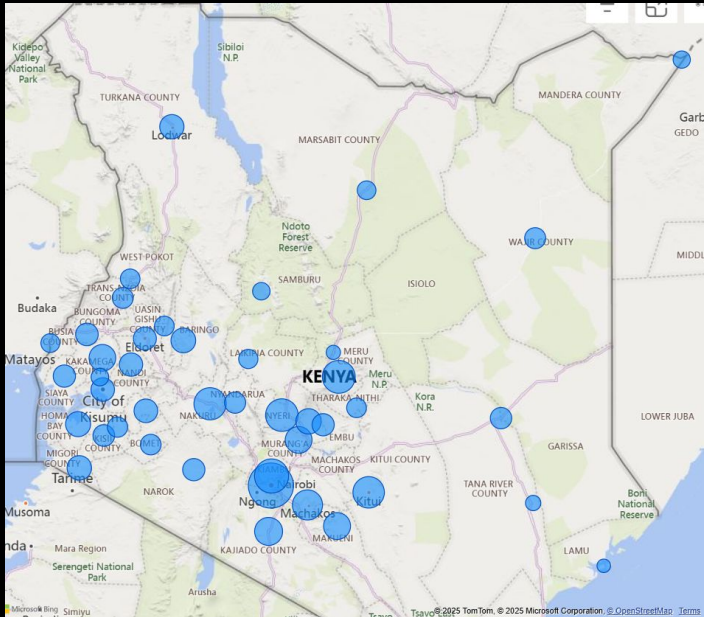
1. Standardized column names for consistency.
2. Transformed data into long format using unpivoting.
3. Removed irrelevant rows (e.g., totals, referrals).
4. Cleaned values (trimmed, cleaned, capitalized) and set appropriate data types (text, whole number, dates).
5. Handled missing or decimal values by replacing blanks with 0 and rounding where needed.

Hospital Infrastructure per County

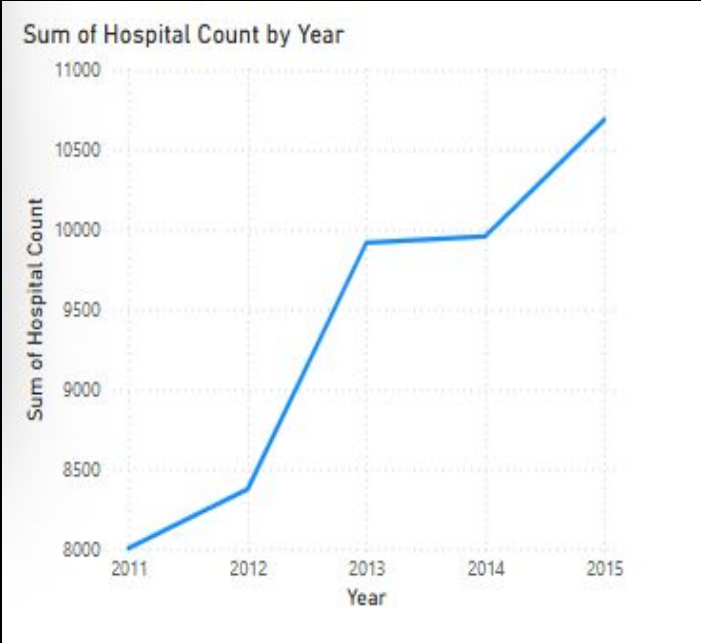


Hospital distribution in the country, rate of Hospital growth , with Nairobi showing major increase compared to other counties, general trend growth of hospitals

Hospital Distribution in Kenya across Counties in 2015

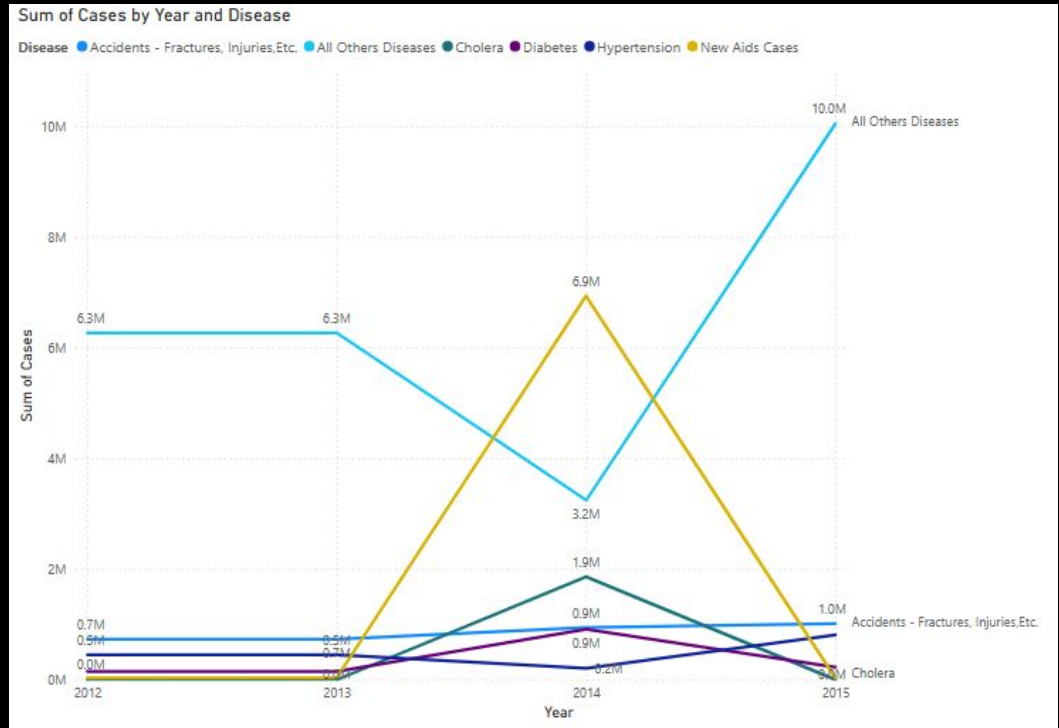
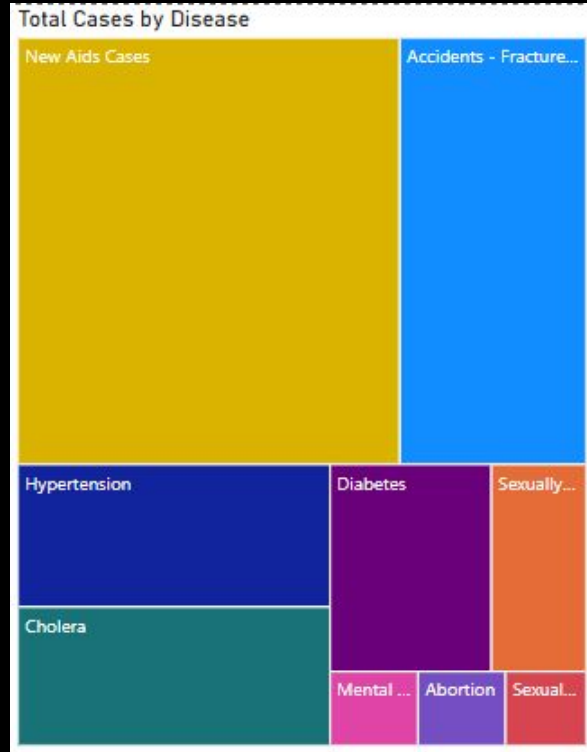


Total Hospitals in 2015
11K



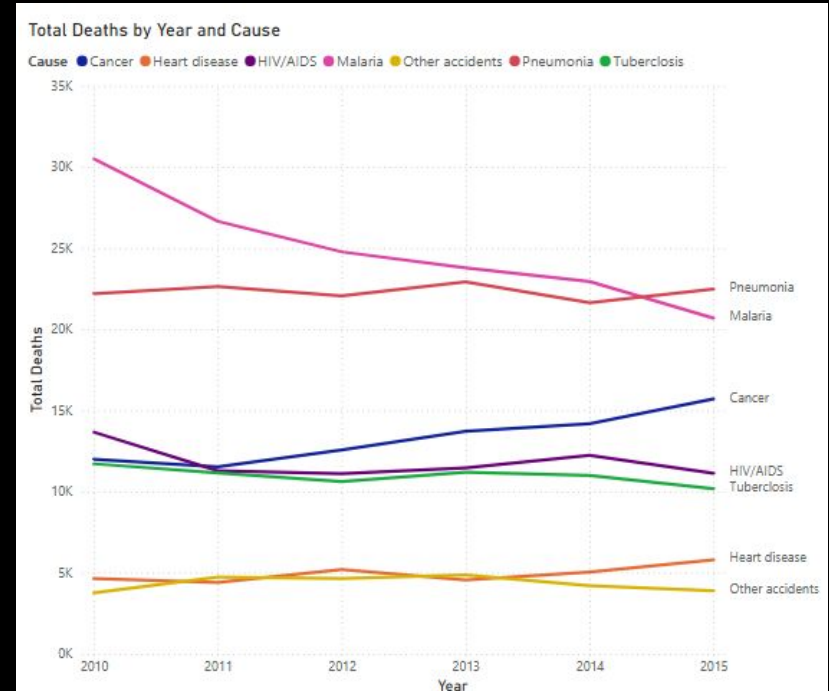
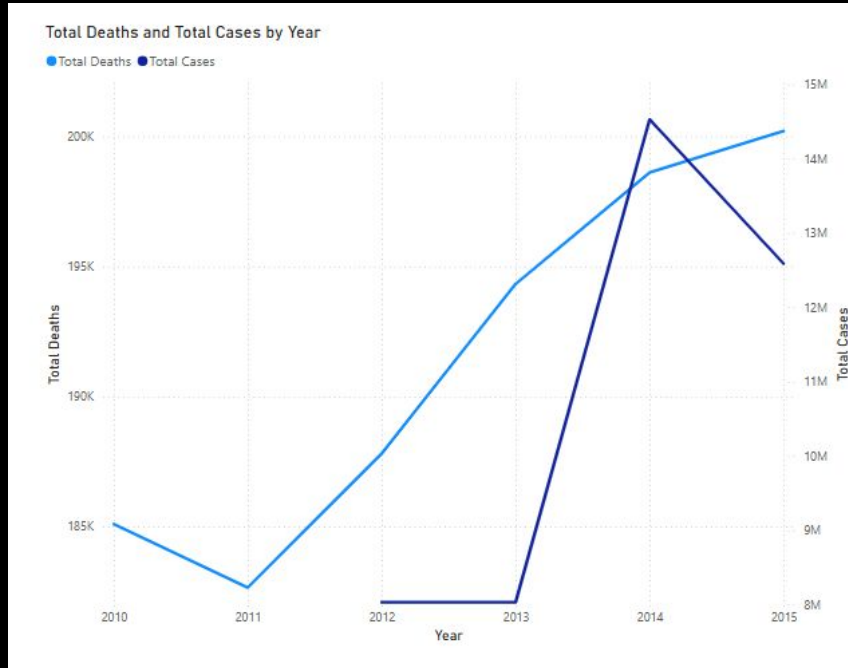
Disease Burden

Total cases of diseases 2011-2015, Trend of diseases over the years



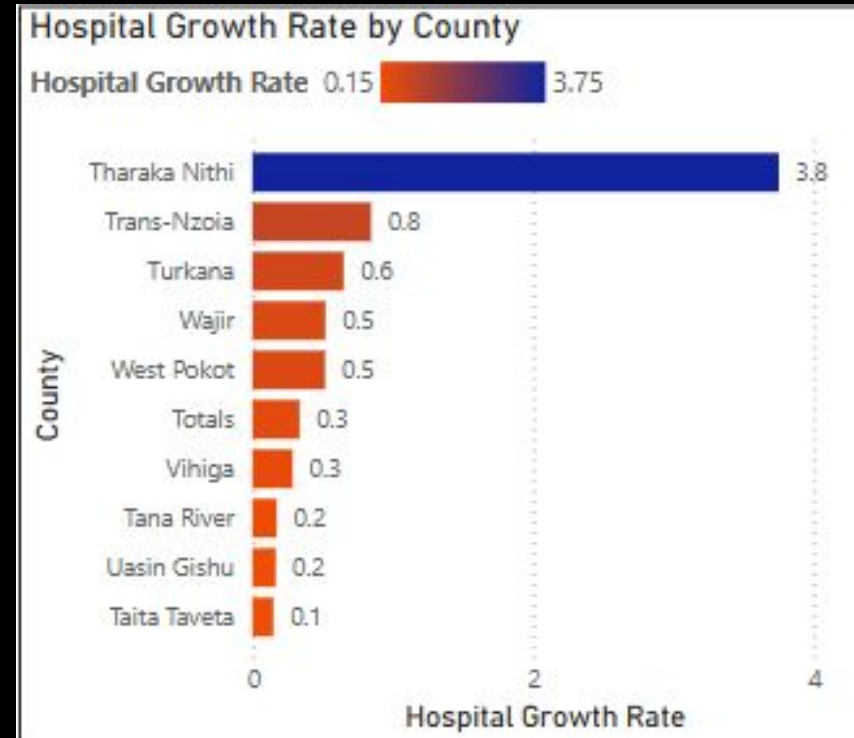
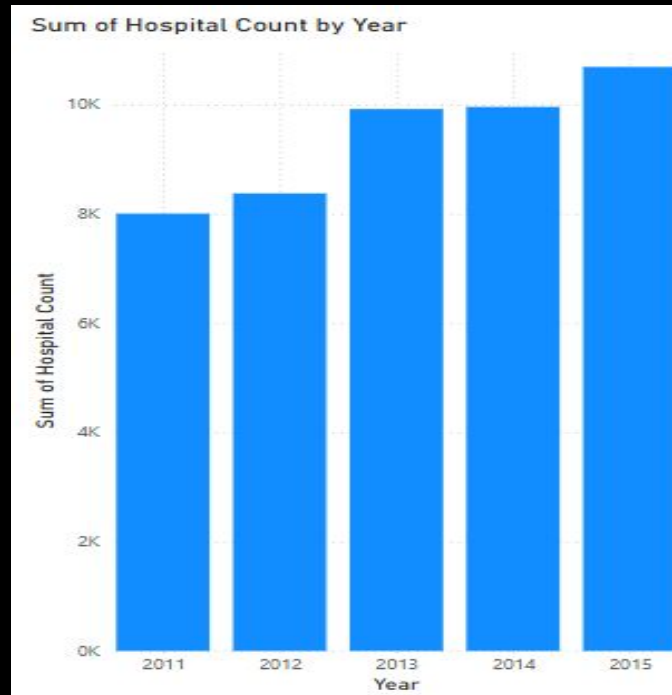
Capacity Vs Cases & Mortality Context

Trends in Disease Incidence and Mortality (2010–2018): Left chart displays the upward trends in total disease cases and deaths, while the right chart breaks down mortality by causes such as cancer and heart disease, illustrating regional variations over time



Hospital growth over years & Counties

Hospital Growth Over the Years and by County: The bar chart on the left illustrates the increasing hospital count by year, while the chart on the right highlights varying growth rates across different counties, identifying those with the highest hospital expansion.





Key National Trends

Insight into Data



- Despite the spike in cases, the death rate from AIDS specifically wasn't proportionally high — suggesting treatment access or improved care may have mitigated fatality
- Communicable diseases continue to dominate Kenya's disease burden. This suggests ongoing public health challenges in areas like sanitation, vector control, and vaccination.
- Counties with the highest number of hospitals also report the highest disease burden.
- Why doesn't an increase in hospitals lead to a visible reduction in disease burden? Is hospital care too reactive instead of preventive?
- The rate of hospital growth continues to increase over time reflecting ongoing investment in healthcare infrastructure.
- There was a notable spike in deaths in 2014, partly due to a rise in new AIDS cases but also a surge in terror attacks that likely contributed to the overall mortality rate.



Limitations

- **Reporting Bias:** Reported disease cases may not reflect true incidence due to factors like healthcare access, reporting practices, and diagnostic capacity varying by county.
- **Hospital Metrics:** Hospital *count* is a proxy for capacity; actual bed numbers, staffing levels, and service availability would provide a more accurate measure.

With the Data

- **Disease Data Integrity (2014, 2013):** The 2014 disease data is largely unusable as presented, and 2013 data is suspect as it Largely duplicates 2012 data for many entries, raising questions about its accuracy or reporting method for that year.
- **Deaths Data Granularity:** Lack of county-level death data prevents direct analysis of how county-level factors (hospitals, specific disease cases) correlate with mortality outcomes *within that county*.

What next?



For the Government & Health Policymakers

Strengthen Preventive Healthcare Systems

Despite more hospitals, disease burden remains high — pointing to the need for stronger preventive health strategies such as community health education, sanitation programs, and routine screenings.

Target High-Burden Counties with Tailored Interventions

High disease cases in counties with many hospitals suggest that access alone isn't enough. Allocate resources to address root causes like poor living conditions, overcrowding, and environmental risks.

Track and Improve AIDS Management Programs

The low fatality rate despite high AIDS incidence hints at effective treatment programs. Scale up these success stories to counties with poor health outcomes.

What next?



For the Public Health Sector

Invest More in Community-Based Care

Shift focus from facility-based, reactive care to proactive, preventive outreach — including mobile clinics, home visits, and disease awareness campaigns.

Enhance Disease Surveillance and Reporting Systems

High case numbers in well-equipped counties may reflect better reporting. Expand these systems to underreporting areas for a more accurate national picture.

Scale Vaccination & Sanitation Efforts

Ongoing burden from communicable diseases points to a need for renewed investment in immunization, water safety, and hygiene infrastructure.

Conclusion



Long-term progress depends on balancing **investment in care** with **prevention, education, and equity**.

THANKYOU!